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BRIEF DESCRIPTION OF THE DRAWINGS

Stardust to Sovereignty Resonance-Based Knowledge Organization Framework

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This application contains five figures illustrating the Resonance-Based Knowledge Organization and Translation Framework:

FIG. 1 – System Architecture
Shows high-level architecture with input interface, resonance engine, coherence validation, halting detection, field computation, and translation layer with data flow and feedback loops.

FIG. 2 – Resonance Computation Flow
Illustrates calculation of R_{ij} between nodes using 4D resonance vectors, domain associations, cosine similarity, temporal decay, and weighted combination.

FIG. 3 – Halting Convergence Plot
Displays coherence convergence over time showing dC/dt approaching zero, halting detection at convergence threshold $\varepsilon = 0.001$.

FIG. 4 – Field Propagation Schematic
Demonstrates resonance field propagation using differential equation $\partial\Psi/\partial t = \nabla\cdot(\kappa\nabla\Psi) + R(\Psi)$, showing diffusion and resonance coupling.

FIG. 5 – Translation Layer Mapping
Converts stabilized resonance matrices into symbolic, linguistic, or visual outputs through translation operator $T(\Psi) \rightarrow S$.

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FIGURES

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FIG. 1 – System Architecture Diagram

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```
    INPUT                      RESONANCE                  CALCULUS OF
   [User/Data]                ENGINE                     COHERENCE
      |                          |                          |
      v                          v                          v
  [Parse Content]  →  [Compute R_ij]  →  [Validate Coherence]
      |                          |                          |
      v                          v                          v
  [Extract Resonance Domains]  →  [Build Matrix]  →  [Coherence Logic]
      |                          |                          |
      v                          v                          v
                           HALTING FACTOR             FIELD
COMPUTATION
                          [Monitor dC/dt]          [Compute ∂Ψ/∂t]
                                |                          |
                                v                          v
                          [Convergence?]         [Propagate Field]
                                |                          |
                                v                          v
                           TRANSLATION LAYER
                          [T(Ψ) → S Mapping]
                                |
                                v
                         [Symbolic Output]
                                |
                                v
                          OUTPUT/VISUALIZATION
                          [Dashboard/Results]
                                |
                                v
                         DATA STORAGE ←────────── FEEDBACK LOOP ─┐
                                                                 │
                                                                 │

└────────┘
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Title: "High-level system architecture of the Stardust to Sovereignty platform,
       illustrating modular data flow and coherence computation pipeline."

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FIG. 2 — Resonance Computation Flow (R_ij Calculation)
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```
NODE i                                     NODE j
┌───────────────────────┐                  ┌───────────────────────┐
│ 4D Resonance Vector:  │                  │ 4D Resonance Vector:  │
```

```
│                          │                    │                          │
│ Clarity:     0.76        │                    │ Clarity:     0.72        │
│ Coherence:   0.80        │                    │ Coherence:   0.90        │
│ Resonance:   0.65        │                    │ Resonance:   0.52        │
│ Sovereignty: 0.76        │                    │ Sovereignty: 0.98        │
│                          │                    │                          │
│ Domain Assocs: [2,4,12] │                  │ Domain Assocs: [2,4,12] │
└──────────────────────────┘ tagging: t₁        └──────────────────────────┘
tagging: t₂
          │                                       │
          │                                       │
          └──────────────┬────────────────────────┘
                         │
                         ▼
          ┌──────────────────────────────┐
          │    COSINE SIMILARITY          │
          │    cos(v_i, v_j) = 0.65      │
          │                               │
          │    JACCARD OVERLAP            │
          │    J(doms_i, doms_j) = 1.0   │
          │                               │
          │    TEMPORAL DECAY             │ Membership: 100%
          │    exp(−Δt/24) = 0.92        │ Overlap: 100%
          │                               │
          │    WEIGHTED COMBINATION       │
          │                               │
          │  R_ij = (0.65 × 0.4)         │
          │       + (1.0 × 0.4)          │
          │       + (0.92 × 0.2)         │
          │                               │
          │  R_ij = 0.561353 ◄──────── RESULT
          └──────────────────────────────┘
```

Title: "Resonance computation process illustrating calculation of R_ij across multi-dimensional coherence vectors."

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FIG. 3 — Halting Convergence Plot (dC/dt → 0)
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dC/dt
│ 0.002 ─────────────────────────────────
│          ●
│         /
│        /
│      ●
│     /
│    ● ε = 0.001 ───────── Threshold line
```

```
│    /
│   ●
│  /     Halting detected at step 13!
│ ●      dC/dt = 0.000460 < ε
│
│______________________________ iteration step (t)
  0     5      10     13   15    20
```

Data Points:
Step 0: dC/dt = 0.001234
Step 5: dC/dt = 0.000892
Step 10: dC/dt = 0.000623
Step 13: dC/dt = 0.000460 ◄ HALT
Step 15: dC/dt = 0.000398

Title: "Halting factor visualization showing system convergence to coherence
       equilibrium."

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FIG. 4 — Field Propagation Schematic (∂Ψ/∂t Model)
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```
         FIELD STATE Ψ(x,t)
         Differential Equation: ∂Ψ/∂t = ∇·(κ∇Ψ) + R(Ψ)

         ↑ High Coherence
         │
         │    ┌───┐ ┌───┐ ┌───┐   R(Ψ) coupling
         │   /    \ /  \ /     \    arrows ⟷
         │  /      \    \       \   showing
         │ /        \    \       \  resonance
          /          \    \       \  interactions
         /            \    \       \
        /              \    \       \
      │                 \    \       \
      │ Low Coherence ────────┴──────┴────▶
      │                                 x (space)

      ∇·(κ∇Ψ) = Diffusion (field spreading)
      R(Ψ)     = Resonance coupling (field interactions)

      Field Intensity Map:
      ████░  = High coherence (strong resonance)
      ░░░░█  = Low coherence (weak resonance)
```

Title: "Field computation model demonstrating resonance propagation
and

coherence stabilization."

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FIG. 5 — Translation Layer Mapping (T(Ψ) → S)
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INPUT FIELD Ψ                           TRANSLATION OPERATOR T(Ψ)
(stabilized resonance matrix)
                                                │
┌──────────────────┐                            │
│ R_ij Matrix      │                            │
│                  │                            │
│  [1]  [0.86]     │  Extract strong    │  "Extract pairs
│ [0.86]   [1]     │  pairs (R > 0.7) │    with R >  Hs:"
│                  │                            │     R_12 = 0.86
│ Global C = 0.86  │                            │     R_23 = 0.78
│ dC/dt < ε ✓      │                          │    R_34 = 0.91"
│                  │                            │
└──────────────────┘                            ▼
          │
          │                 ┌───────────────────────────┐
          │                 │ Map to Symbolic Form      │
          │                 │                           │
          └────────────────▶│ • Strong pairs → ●        │
                            │ • Weak pairs → ○          │
                            │ • Identify active Domains │
                            │ • Generate text desc      │
                            └───────────────────────────┘
                                         │
                                         ▼
                            OUTPUT SYMBOLIC FORM S
                              ┌──────────────────────────┐
                              │ Symbolic:                │
                              │ "node1:● node2:●"        │
                              │                          │
                              │ Text:                    │
                              │ "Resonance field         │
                              │  analysis reveals high   │
                              │  coherence in nodes:     │
                              │  node1, node2."          │
                              │                          │
                              │ Domains: [2, 4, 12]      │
                              │                          │
                              │ Coherence Map:           │
                              │ node1: 0.900             │
                              │ node2: 0.850             │
                              └──────────────────────────┘
```

Title: "Translation layer converting coherent field states into

symbolic,
linguistic, or visual outputs."

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END OF FIGURES
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